WORKED SOLUTION

STEP I 1992, Question 7

Pure • Integer-Valued Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q7

The key idea

Don't try to prove that the coefficients are integers, because for the quadratic they needn't be: $\frac{1}{2}x^2 + \frac{1}{2}x$ takes integer values at every integer. Instead, express the coefficients in terms of the given integer values and substitute back, so that $\mathrm{f}(n)$ appears as a combination of $\mathrm{f}(-1)$, $\mathrm{f}(0)$ and $\mathrm{f}(1)$ with integer multipliers. The final part is then just the middle part again, after a shift of origin.

Solution

The linear function

Since $\mathrm{g}(0) = b$ and $\mathrm{g}(1) = a + b$, we have $b = \mathrm{g}(0)$ and $a = \mathrm{g}(1) - \mathrm{g}(0)$, and both are integers. So for any integer n,

$$\mathrm{g}(n) = an + b = \mathrm{g}(0) + n(\mathrm{g}(1) - \mathrm{g}(0)),$$

which is a sum of products of integers, so it's an integer.

The quadratic function

Write $p = \mathrm{f}(-1)$, $q = \mathrm{f}(0)$, $r = \mathrm{f}(1)$, all integers. Then

$$p = A - B + C, \qquad q = C, \qquad r = A + B + C,$$

and solving,

$$C = q, \qquad B = \frac{r-p}{2}, \qquad A = \frac{p+r-2q}{2}.$$

So A and B need not be integers; they could be halves of odd integers. Substituting these into $\mathrm{f}(n) = An^2 + Bn + C$ and collecting the terms in p, q and r:

$$\begin{aligned}\mathrm{f}(n) &= \frac{p+r-2q}{2}\,n^2 + \frac{r-p}{2}\,n + q \\ &= \frac{n(n-1)}{2}\,p + (1-n^2)\,q + \frac{n(n+1)}{2}\,r.\end{aligned}$$

Ask yourself The multipliers of p and r have a 2 in the denominator. Why are they integers?

For an integer n, the numbers $n-1$ and n are consecutive integers, so one of them is even, and $n(n-1)$ is even. Likewise $n(n+1)$ is even. So $\frac{1}{2}n(n-1)$ and $\frac{1}{2}n(n+1)$ are integers, as is $1-n^2$. Hence $\mathrm{f}(n)$ is a sum of products of integers, and so it's an integer for every integer n, as required.

(The three multipliers are the quadratics that equal 1 at one of $-1, 0, 1$ and 0 at the other two. This is Lagrange interpolation.)

The final part

Let α be real and define $\mathrm{h}(x) = \mathrm{f}(\alpha + x)$. Expanding,

$$\mathrm{h}(x) = A(\alpha+x)^2 + B(\alpha+x) + C = Ax^2 + (2A\alpha + B)x + (A\alpha^2 + B\alpha + C),$$

so h is also a function of the form $A'x^2 + B'x + C'$ with real coefficients. The middle part used nothing about A, B, C except that they were real numbers. Now

$$\mathrm{h}(-1) = \mathrm{f}(\alpha - 1), \qquad \mathrm{h}(0) = \mathrm{f}(\alpha), \qquad \mathrm{h}(1) = \mathrm{f}(\alpha + 1)$$

are integers by hypothesis. So, by the middle part applied to h, $\mathrm{h}(n) = \mathrm{f}(\alpha + n)$ is an integer for every integer n. Explicitly,

$$\mathrm{f}(\alpha+n) = \frac{n(n-1)}{2}\,\mathrm{f}(\alpha-1) + (1-n^2)\,\mathrm{f}(\alpha) + \frac{n(n+1)}{2}\,\mathrm{f}(\alpha+1).$$

Checks

- Put $n = -1, 0, 1$ into the formula for $\mathrm{f}(n)$: the multipliers become $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$, giving back p, q, r.
- The example $\mathrm{f}(x) = \frac{1}{2}x^2 + \frac{1}{2}x$ has $p = 0$, $q = 0$, $r = 1$, and the formula gives $\mathrm{f}(n) = \frac{1}{2}n(n+1)$, the triangular numbers.
- Numerically, with $\alpha = \sqrt{2}$ and $\mathrm{f}(\alpha - 1) = 3$, $\mathrm{f}(\alpha) = -7$, $\mathrm{f}(\alpha + 1) = 11$, the quadratic through these points gives $\mathrm{f}(\alpha + 5) = 363.000\ldots$, matching $10 \cdot 3 - 24 \cdot (-7) + 15 \cdot 11 = 363$.

Common slips

- Claiming that A, B, C must be integers. They needn't be, and an argument that relies on it is wrong.
- In the final part, trying to show that α must be an integer or rational. It can be any real number; only the shift matters.

Beyond STEP

Polynomials that take integer values at every integer are exactly the integer combinations of the binomial coefficients $\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$. For a quadratic these are 1, x and $\frac{1}{2}x(x-1)$, and the result is proved with finite differences: a polynomial of degree d is integer-valued if it takes integer values at $d+1$ consecutive integers. The same idea underlies Newton's forward-difference formula in numerical analysis. For more on integer-valued functions and divisibility, see Davenport, *The Higher Arithmetic*.

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